

SECTION 5.2: SERIES (DAY 1)

Things to know by the end of this section

- a. how to use sigma notation *with facility*
- b. the meaning of a *series*, especially as compared to a *sequence* (from §5.1)
- c. the meaning of a *sequence of partial sums of a series* and how to find it.
- d. what it means to say a series converges.
- e. what a *geometric series* is and how to determine whether or not it converges.
- f. what a *telescoping series* is and how to determine whether or not it converges.

1. An infinite series is *a sum of infinitely many terms.*

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots + a_n + \dots$$

$\uparrow$ 1st term $\uparrow$ 2nd term $\uparrow$ nth term

Series

$a_1, a_2, a_3, \dots$ ← terms of the series

2. The sequence of partial sums of a series is $\{S_n\}_{n=1}^{\infty}$ where

$$\begin{aligned}
 S_1 &= a_1 \\
 S_2 &= a_1 + a_2 \\
 S_3 &= a_1 + a_2 + a_3 \\
 &\vdots \\
 S_n &= a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n
 \end{aligned}$$

$S_1, S_2, S_3, S_4, \dots$ *Sequence*

3. A starter example

• $\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots + \frac{1}{2^n} + \dots$ ← Series

• terms in the series: $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots, \frac{1}{2^n}, \dots$ ← Sequence

• Sequence of partial sums:

$$\begin{aligned}
 S_1 &= \frac{1}{2} \\
 S_2 &= \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \\
 S_3 &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8} \\
 S_4 &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{15}{16}
 \end{aligned}$$

or
 $0.5, 0.75, 0.875, 0.9375, 0.96875, \dots$
 $S_1 \quad S_2 \quad S_3 \quad S_4 \quad S_5$

4. For each series below,

- expand the sigma notation,
- write the first 5 terms in the series,
- write the first 5 terms in the sequence of partial sums, S_1, S_2, S_3, S_4, S_5 . (Use a calculating device to get a decimal or fraction representation of the partial sums.)

(a) $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \textcircled{i} \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \frac{16}{81} + \frac{32}{243} + \dots$

$\textcircled{ii} \frac{2}{3}, \frac{4}{9}, \frac{8}{27}, \frac{16}{81}, \frac{32}{243} \leftarrow \text{less than 1}$

$\textcircled{iii} S_1 = \frac{2}{3} = \underline{0.\bar{6}}$

$S_2 = \frac{2}{3} + \frac{4}{9} = \underline{1.\bar{1}}$

$S_3 = \frac{2}{3} + \frac{4}{9} + \frac{8}{27} = \underline{1.\bar{407}}$

$S_4 = \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \frac{16}{81} = \underline{1.60493}$

$S_5 = \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \frac{16}{81} + \frac{32}{243} = \underline{1.7366}$
 $\uparrow$ more than one

$\{S_n\} = 0.\bar{6}, 1.\bar{1}, 1.\bar{407}, 1.6049, 1.736, \dots$

● Maybe convergence?
(It does!)

(b) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \textcircled{c} \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} + \dots$

$\textcircled{ii} \frac{1}{2}, \frac{1}{6}, \frac{1}{12}, \frac{1}{20}, \frac{1}{30}$

$\textcircled{iii} S_1 = \frac{1}{2}$

$S_2 = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$

$S_3 = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} = \frac{3}{4}$

$S_4 = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} = \frac{4}{5}$

$S_5 = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} = \frac{5}{6}$

$\{S_n\} = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$

● Guess convergence
 $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$
 $\uparrow$
 pattern recognition.

$$(c) \sum_{n=1}^{\infty} \frac{(-1)^n}{5} \stackrel{(i)}{=} -\frac{1}{5} + \frac{1}{5} - \frac{1}{5} + \frac{1}{5} - \frac{1}{5} + \dots$$

$$(ii) \text{ terms of series } -\frac{1}{5}, \frac{1}{5}, -\frac{1}{5}, \frac{1}{5}, -\frac{1}{5}$$

$$(iii) S_1 = -\frac{1}{5} \quad \{S_n\} = -\frac{1}{5}, 0, -\frac{1}{5}, 0, -\frac{1}{5}, \dots$$

$$S_2 = -\frac{1}{5} + \frac{1}{5} = 0$$

$$S_3 = -\frac{1}{5} + \frac{1}{5} - \frac{1}{5} = -\frac{1}{5}$$

$$S_4 = 0$$

$$S_5 = -\frac{1}{5}$$

● Definitely diverges.
 $\lim_{n \rightarrow \infty} S_n = DNE$

$$(d) \sum_{n=1}^{\infty} \frac{n}{n^2 + 2} \stackrel{(i)}{=} \frac{1}{1+2} + \frac{2}{4+2} + \frac{3}{9+2} + \frac{4}{16+2} + \frac{5}{25+2} + \dots$$

$$(ii) \frac{1}{3}, \frac{2}{6}, \frac{3}{11}, \frac{4}{18}, \frac{5}{27}$$

● Maybe converges?
 (It doesn't)

$$(iii) S_1 = \frac{1}{3}$$

$$S_2 = \frac{1}{3} + \frac{2}{6} = \frac{2}{3}$$

$$S_3 = \frac{1}{3} + \frac{2}{6} + \frac{3}{11} = 0.\overline{93}$$

$$S_4 = \frac{1}{3} + \frac{2}{6} + \frac{3}{11} + \frac{4}{18} = 1.\overline{16}$$

$$S_5 = \frac{1}{3} + \frac{2}{6} + \frac{3}{11} + \frac{4}{18} + \frac{5}{27} = 1.346$$

$$EVI \quad S_6 = 1.5046, S_7 = 1.6419 \dots$$

5. **Definition:** Given the series

$\sum_{n=1}^{\infty} a_n$, its sequence of partial sums is $\{S_n\}_{n=1}^{\infty}$

• The series converges if

$\lim_{n \rightarrow \infty} S_n = L$ (ie S_n converges)

• The series diverges if

$\lim_{n \rightarrow \infty} S_n$ does not exist (ie S_n diverges)

6. Revisit the series in #4. Which ones do you think converge? Diverge? How sure are you about your conclusions?